

Homework for Chap 13

Given a direct bandgap semiconductor GaAs with a bandgap energy $E_g = 1.42$ eV at room temperature (300 K), its absorption coefficient satisfies the relationship with the incident photon energy E : when $E \geq E_g$, $\alpha \propto (E - E_g)^{1/2}$. When $E < E_g$, $\alpha \approx 10$ cm⁻¹ (negligible), here E is the photon energy. A monochromatic light is normally incident on a thin GaAs slab with the following conditions:

- Incident light wavelength $\lambda_1 = 1000$ nm;
- Incident light wavelength $\lambda_2 = 800$ nm;

Suppose the thickness of the GaAs slab $d = 1$ μ m, the Reflectivity of the semiconductor surface $R = 0.3$ for both λ_1 and λ_2 , and neglect multiple reflections:

1. Determine whether the incident light with wavelengths λ_1 and λ_2 can be effectively absorbed by the GaAs semiconductor, respectively, and explain the reasons.
2. If the absorption coefficient corresponding to the incident light of wavelength λ_1 and λ_2 is $\alpha = 10$ cm⁻¹ and $\alpha = 1 \times 10^4$ cm⁻¹, respectively, calculate the transmittance T of the light after passing through the GaAs slab for light with λ_1 and λ_2 .
3. In a photoluminescence spectrum measurement, a Ti:sapphire continuous wave (cw) laser delivers monochromatic light with wavelength $\lambda_2 = 800$ nm and the laser output power is 10 mW. If the laser beam is focused uniformly onto a round area with a radius $r = 100$ μ m. Given the $R = 0.3$, $\alpha = 1 \times 10^4$ cm⁻¹, the photogenerated carrier lifetime $\tau = 1$ ns, and suppose the photogenerated carriers are confined in the cylinder with the height $h = 1$ μ m, and the radius of top base and bottom base $r = 100$ μ m, calculate the average density of the photogenerated electrons.

HW For Ch 13

陈信珂 12311125

1. $E_1 = \frac{hc}{\lambda_1} = 1.24 \text{ eV} < E_g$ 不能被有效吸收

$E_2 = \frac{hc}{\lambda_2} = 1.55 \text{ eV} > E_g$ 可以被有效吸收

2. ① $\lambda_1 = 1000 \text{ nm}$, $\alpha = 10 \text{ cm}^{-1}$

$T_1 = (1-R) e^{-\alpha d} = 69.9\%$

② $\lambda_2 = 800 \text{ nm}$, $\alpha = 10^4 \text{ cm}^{-1}$

$T_2 = (1-R) e^{-\alpha d} = 2.8\%$

3. $E = \frac{hc}{\lambda} = 2.483 \times 10^{-19} \text{ J}$

$\Phi = \frac{P}{E} = 4.03 \times 10^{16} \text{ s}^{-1}$

对于 $R=0.3$, 吸收 $h(1-R)$

$\eta = (1-R)(1 - e^{-\alpha d}) = 0.44$

$G = \eta \cdot \Phi = 1.8 \times 10^{16} \text{ s}^{-1}$

$N = G \cdot \tau = 1.8 \times 10^7$

$n = \frac{N}{V} = 5.7 \times 10^{14} \text{ cm}^{-3}$